



HACETTEPE UNIVERSITY
DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

ELE432
ADVANCED DIGITAL DESIGN

Homework Report - 2

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1. Time Spent

I spent around 5 to 6 hours on this homework. Understanding the FSM state diagram took the most time. I had to go back to the lecture slides and the textbook several times to make sure I got the state transitions right. I also made a few mistakes in the ALU decoder at first and had to debug them by checking the test vectors one by one.

2. Test Results

I compiled and simulated the controller using Questa. The testbench ran all 40 test vectors and found 0 errors as one can see below in the transcript screenshot.

This screenshot below shows the Questa transcript after running the simulation.



```
VSIM 52> wave zoom range 130ns 210ns
# [130 ns] [210 ns]
VSIM 53> wave zoom range 0ns 100ns
# [0 ns] [100 ns]
VSIM 54> restart -f
# ** Note: (vsim-12125) Error and warning message counts have been reset to '0' because of 'restart'.
# ** Note: (vsim-8009) Loading existing optimized design _opt2
# ** Note: (vsim-12126) Error and warning message counts have been restored: Errors=0, Warnings=1.
# Loading work.testbench(fast)
# Loading work.controller(fast)
# Loading work.mainsm(fast)
# Loading work.aludec(fast)
# Loading work.instrdec(fast)
VSIM 55> run 2000ns
# 40 tests completed with 0 errors
# ** Note: $stop : controller_testbench.sv(81)
# Time: 415 ns Iteration: 1 Instance: /testbench
# Break in Module testbench at controller_testbench.sv line 81
VSIM 56>
```

The transcript also shows that all four modules loaded correctly:

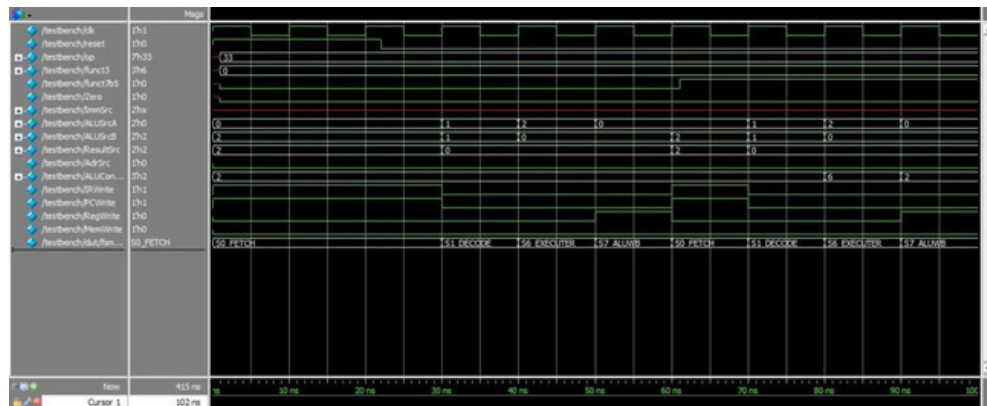
- work.controller
- work.mainsm
- work.aludec
- work.instrdec

Simulation Waveforms

The waveforms below show the controller outputs during simulation.
All signals are in hexadecimal.

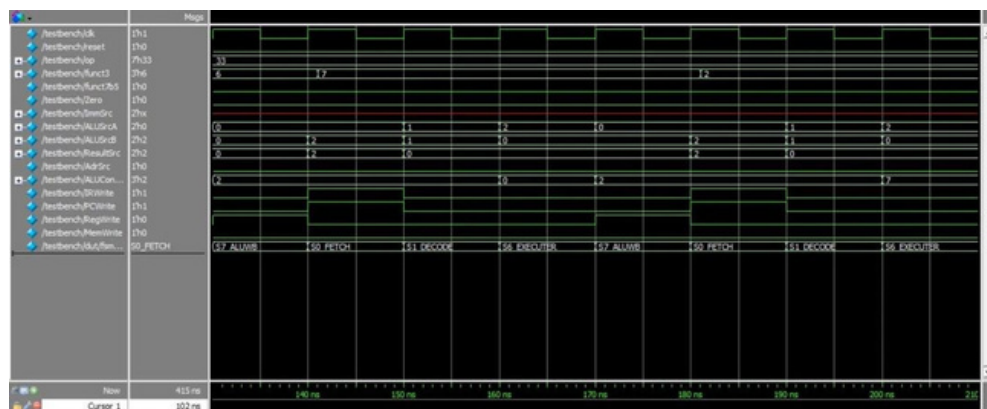
Waveform1 shows two full R-type instruction cycles (0 to 100 ns).

The state sequence is: S0 FETCH -> S1 DECODE -> S6 EXECUTER->S7 ALUWB -> S0 FETCH.
This is the correct 4-cycle sequence for R-type instructions like add, sub, or, and and slt.



Waveform2 shows later cycles (130 to 210 ns) with more R-type instructions.

The same S0 -> S1 -> S6 -> S7 pattern repeats. ALUControl changes based on funct3.
For example, ALUControl = 6 means subtract and ALUControl = 7 means SLT.



Key Signal Observations

Signal	What I observed
IRWrite	Goes high only during S0 FETCH as expected
PCWrite	Goes high during FETCH and JAL states
RegWrite	Goes high only during ALUWB and MemWB states
MemWrite	Goes high only during MemWrite state for sw
ALUSrcA	Changes between 0 1 and 2 depending on state
ALUSrcB	Changes between 0 1 and 2 depending on state
ALUControl	Changes based on instruction type and funct3
state (FSM)	Follows the correct sequence for each instruction type